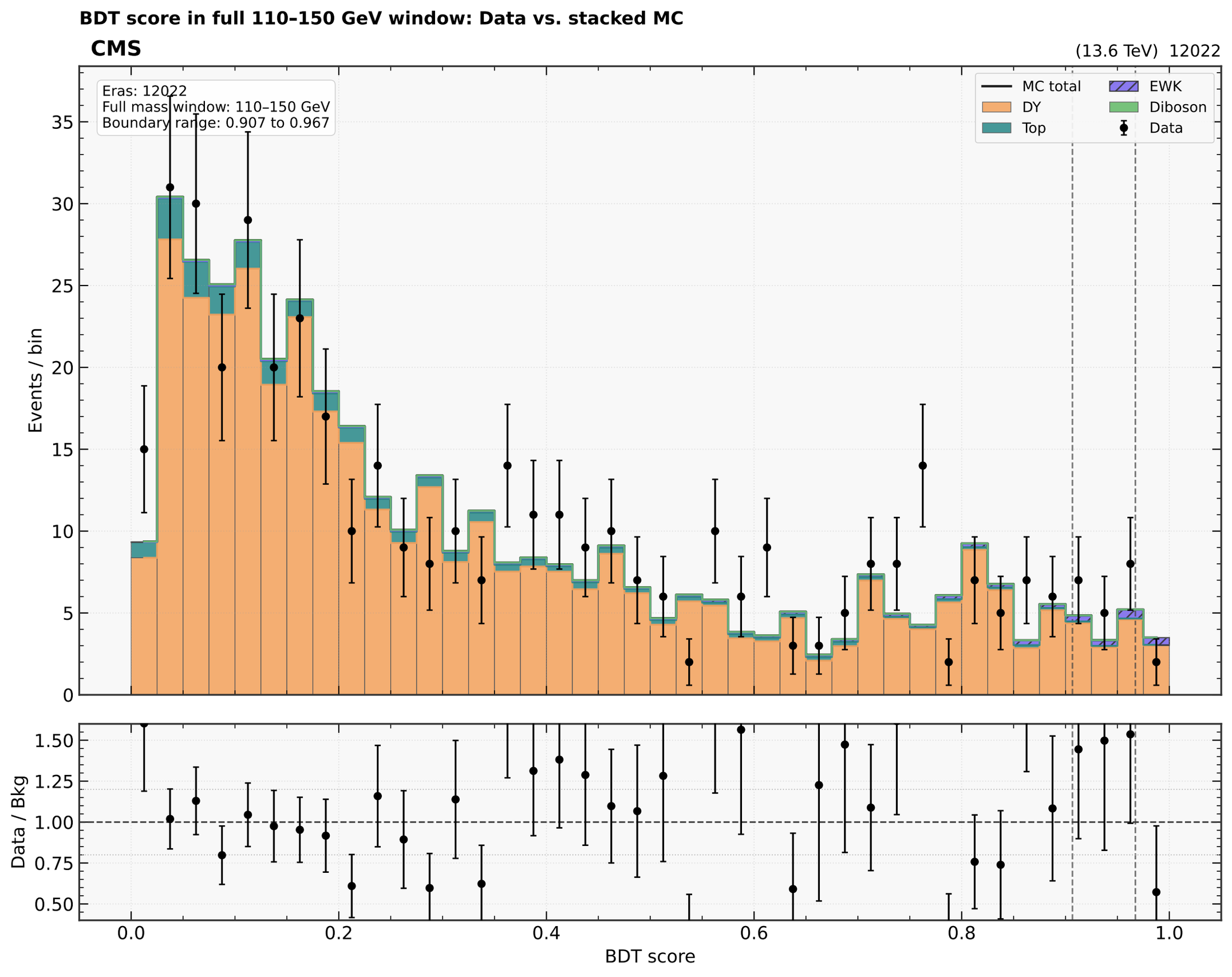
# Run-3 H → μμ VBF BDT Category Optimisation

Eras : 12022
Category edges : 0.0000, 0.9067, 0.9274, 0.9673, 1.0000
MC scale factor : 1.0000 (Data/MC sideband normalisation)
Blinding window : 115–135 GeV
Sidebands : (110,115) ∪ (135,150) GeV
Outputs dir : /eos/home-m/memmad/HiggstoDimuon_VBF/src/drivers/9_TMVA/Outputs
Extra selection : –

CONTENTS:


p.2 BDT score distribution: Data vs. MC
p.3 BDT bin yields
p.4 Significance vs. number of categories
p.5 Boundary scan step-by-step (AN Fig. 32)
p.6 Atanh BDT + S/√B per category
p.7 BDT output vs. mass hypothesis (AN Fig. 26R)
p.8 Signal FWHM vs. BDT score
p.9 ROC curve with AUC
p.10 Blinded dimuon mass spectrum
p.11 Mass sculpting check (AN Fig. 25)
p.12 Per-category mass distributions and background fits
p.13 Background model comparison (χ²/ndf)
p.14 VBF signal DCB fits per category
p.15 DCB parameter table
p.16 σ_CB vs. category
p.17 Signal acceptance × efficiency
p.18 Per-category yield and sensitivity summary
p.19 Signal mass shapes mH=120/125/130 (AN Fig. 40)
p.20 Category summary table
p.21 Category process composition

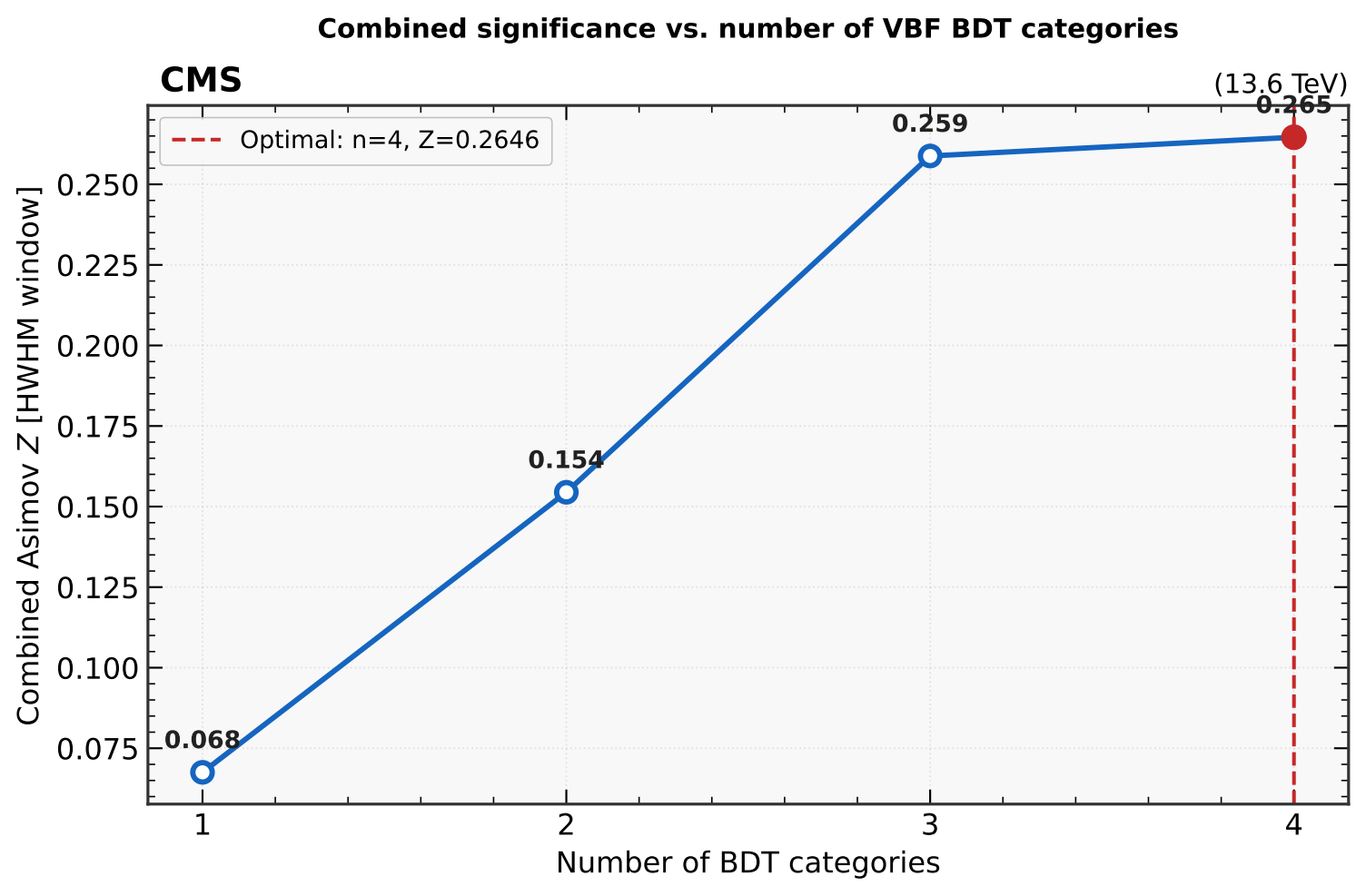

BDT score in full 110–150 GeV window: Data vs. stacked MC
CMS
(13.6 TeV) 12022
Eras: 12022
Full mass window: 110–150 GeV
Boundary range: 0.907 to 0.967
MC total
DY
Top
EWK
Diboson
Data
Events / bin
Data / Bkg
BDT score

# BDT bin summary: signal, background, and sensitivity  (page 1 / 2)

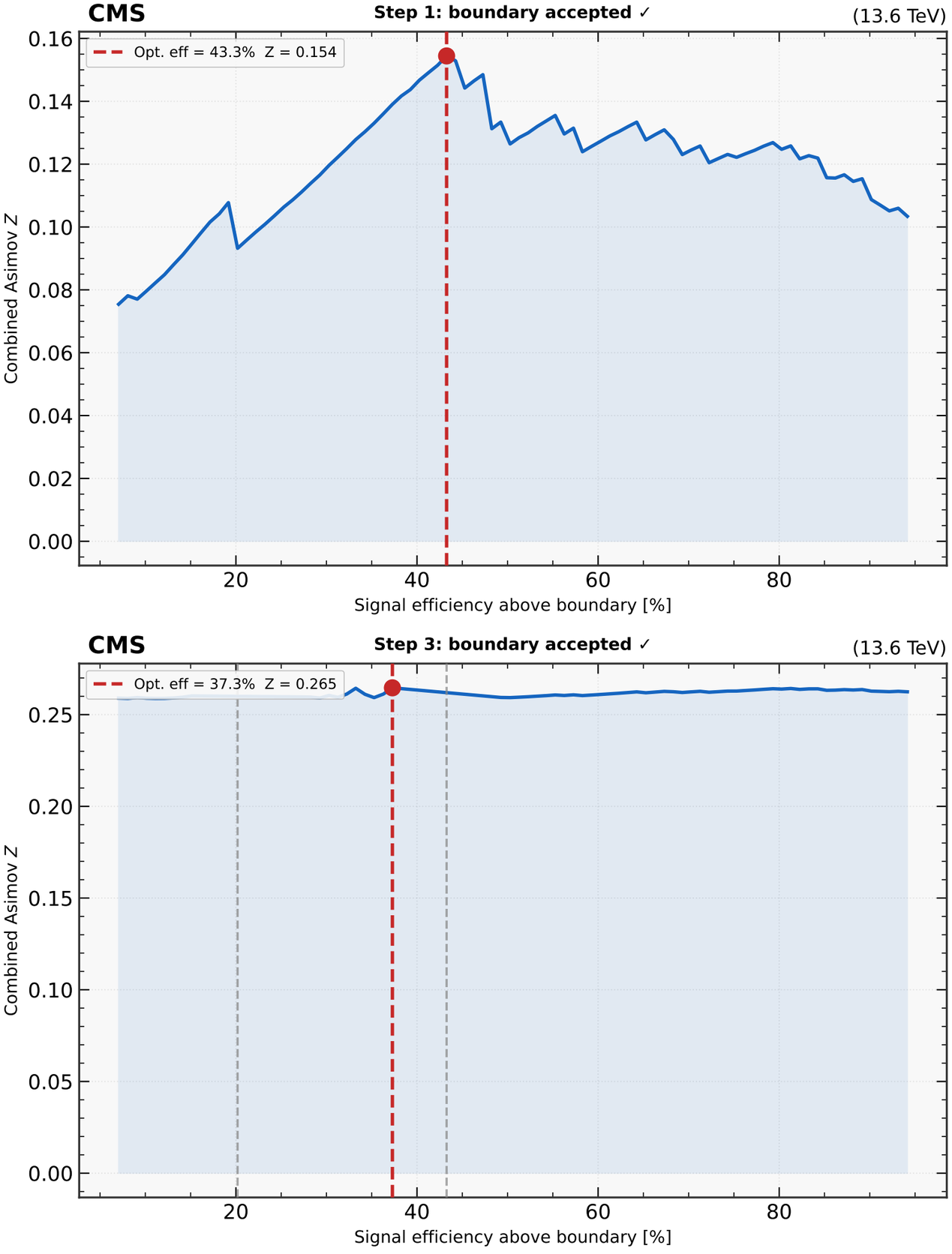
| BDT bin | Total signal | Bkg. ± ΔB | Data | S/(S+B) (%) |
|---|---|---|---|---|
| [0.000, 0.025) | 0.000 | 9.36 ± 1.83 | 15 | 0.00 |
| [0.025, 0.050) | 0.002 | 30.41 ± 3.49 | 31 | 0.01 |
| [0.050, 0.075) | 0.003 | 26.56 ± 3.12 | 30 | 0.01 |
| [0.075, 0.100) | 0.004 | 25.07 ± 3.11 | 20 | 0.01 |
| [0.100, 0.125) | 0.005 | 27.76 ± 3.28 | 29 | 0.02 |
| [0.125, 0.150) | 0.005 | 20.51 ± 2.78 | 20 | 0.03 |
| [0.150, 0.175) | 0.006 | 24.13 ± 3.15 | 23 | 0.02 |
| [0.175, 0.200) | 0.006 | 18.54 ± 2.61 | 17 | 0.03 |
| [0.200, 0.225) | 0.007 | 16.41 ± 2.54 | 10 | 0.04 |
| [0.225, 0.250) | 0.007 | 12.08 ± 2.08 | 14 | 0.06 |
| [0.250, 0.275) | 0.007 | 10.07 ± 1.92 | 9 | 0.07 |
| [0.275, 0.300) | 0.008 | 13.40 ± 2.24 | 8 | 0.06 |
| [0.300, 0.325) | 0.008 | 8.78 ± 1.81 | 10 | 0.09 |
| [0.325, 0.350) | 0.009 | 11.24 ± 2.09 | 7 | 0.08 |
| [0.350, 0.375) | 0.010 | 8.07 ± 1.71 | 14 | 0.12 |
| [0.375, 0.400) | 0.011 | 8.38 ± 1.83 | 11 | 0.13 |
| [0.400, 0.425) | 0.010 | 7.96 ± 1.68 | 11 | 0.12 |
| [0.425, 0.450) | 0.011 | 6.99 ± 1.62 | 9 | 0.16 |
| [0.450, 0.475) | 0.012 | 9.11 ± 1.91 | 10 | 0.13 |
| [0.475, 0.500) | 0.012 | 6.56 ± 1.66 | 7 | 0.19 |

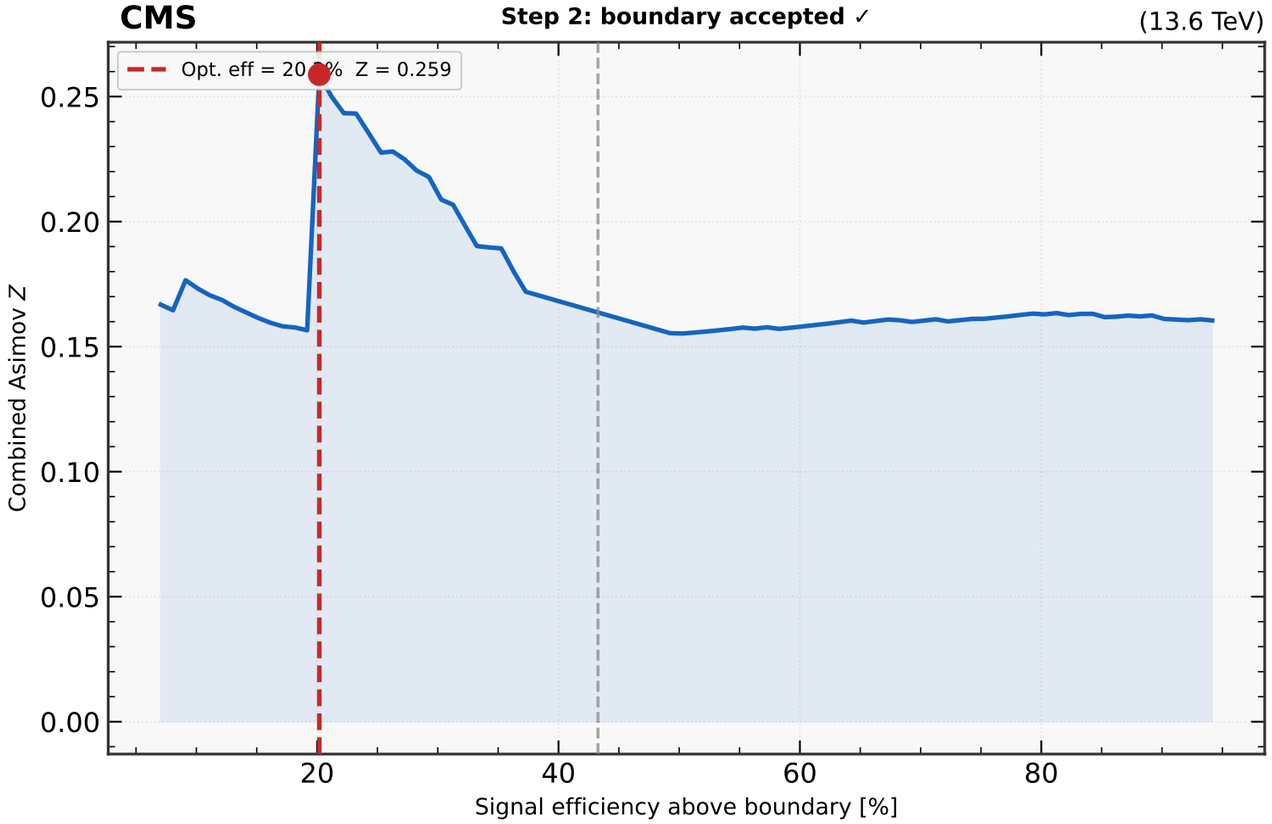
# BDT bin summary: signal, background, and sensitivity  (page 2 / 2)

| BDT bin | Total signal | Bkg. ± ΔB | Data | S/(S+B) (%) |
|---|---|---|---|---|
| [0.500, 0.525) | 0.013 | 4.68 ± 1.12 | 6 | 0.28 |
| [0.525, 0.550) | 0.013 | 6.11 ± 1.63 | 2 | 0.21 |
| [0.550, 0.575) | 0.015 | 5.81 ± 1.47 | 10 | 0.26 |
| [0.575, 0.600) | 0.015 | 3.84 ± 1.07 | 6 | 0.40 |
| [0.600, 0.625) | 0.017 | 3.62 ± 1.05 | 9 | 0.46 |
| [0.625, 0.650) | 0.020 | 5.08 ± 1.38 | 3 | 0.39 |
| [0.650, 0.675) | 0.019 | 2.45 ± 0.85 | 3 | 0.78 |
| [0.675, 0.700) | 0.022 | 3.39 ± 1.15 | 5 | 0.65 |
| [0.700, 0.725) | 0.024 | 7.35 ± 1.72 | 8 | 0.33 |
| [0.725, 0.750) | 0.026 | 4.95 ± 1.35 | 8 | 0.52 |
| [0.750, 0.775) | 0.029 | 4.27 ± 1.32 | 14 | 0.68 |
| [0.775, 0.800) | 0.035 | 6.08 ± 1.47 | 2 | 0.57 |
| [0.800, 0.825) | 0.035 | 9.24 ± 1.92 | 7 | 0.38 |
| [0.825, 0.850) | 0.040 | 6.77 ± 1.61 | 5 | 0.59 |
| [0.850, 0.875) | 0.045 | 3.33 ± 1.02 | 7 | 1.35 |
| [0.875, 0.900) | 0.058 | 5.54 ± 1.46 | 6 | 1.04 |
| [0.900, 0.925) | 0.072 | 4.85 ± 1.32 | 7 | 1.46 |
| [0.925, 0.950) | 0.094 | 3.34 ± 1.10 | 5 | 2.75 |
| [0.950, 0.975) | 0.143 | 5.21 ± 1.44 | 8 | 2.67 |
| [0.975, 1.000) | 0.157 | 3.50 ± 1.18 | 2 | 4.29 |

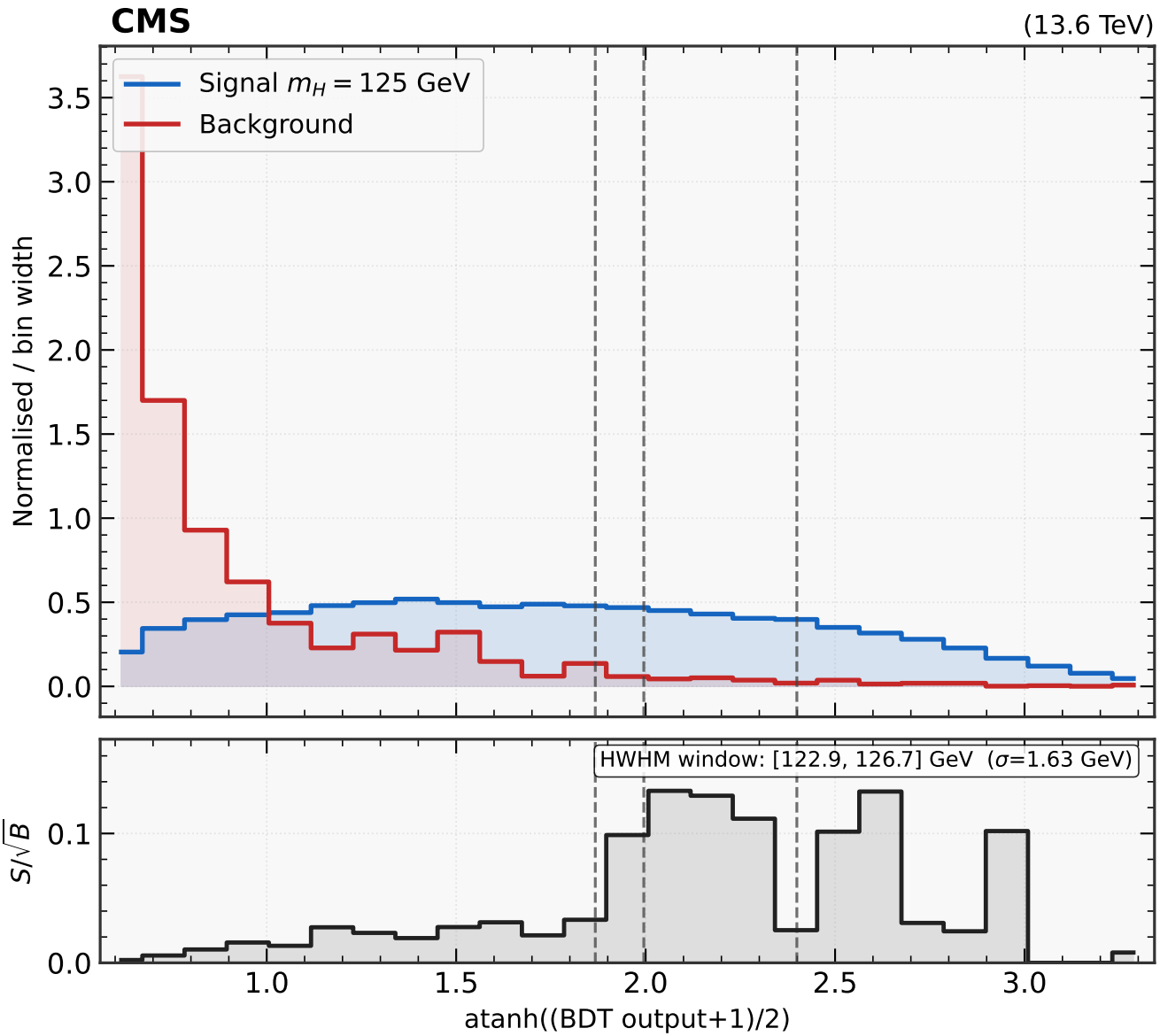

Combined significance vs. number of VBF BDT categories
CMS
(13.6 TeV)
Optimal: n=4, Z=0.2646
0.068
0.154
0.259
0.265
Combined Asimov Z [HWHM window]
0.250
0.225
0.200
0.175
0.150
0.125
0.100
0.075
1
2
3
4
Number of BDT categories

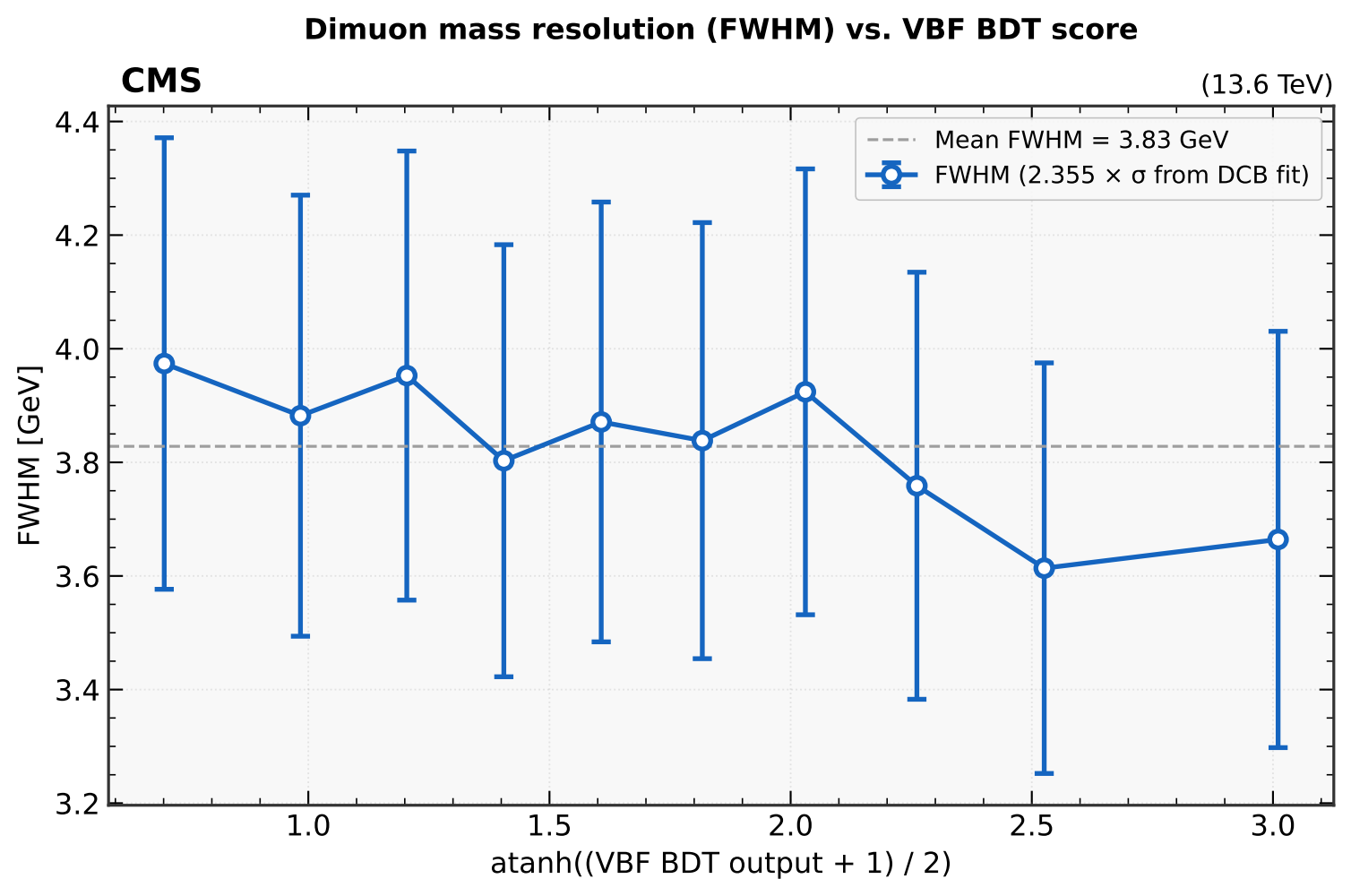
**Iterative category boundary optimisation (AN Fig. 32 analog)**

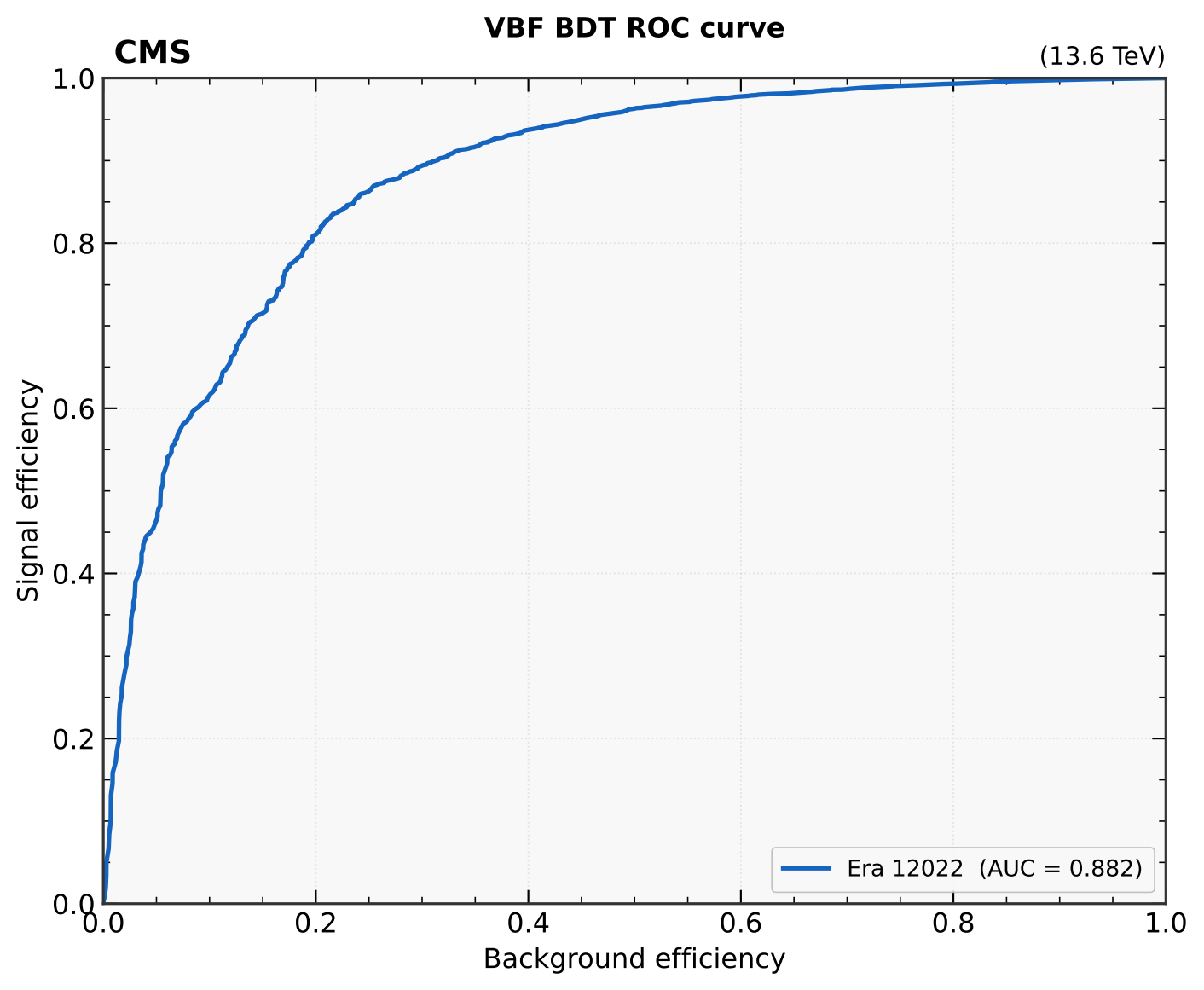


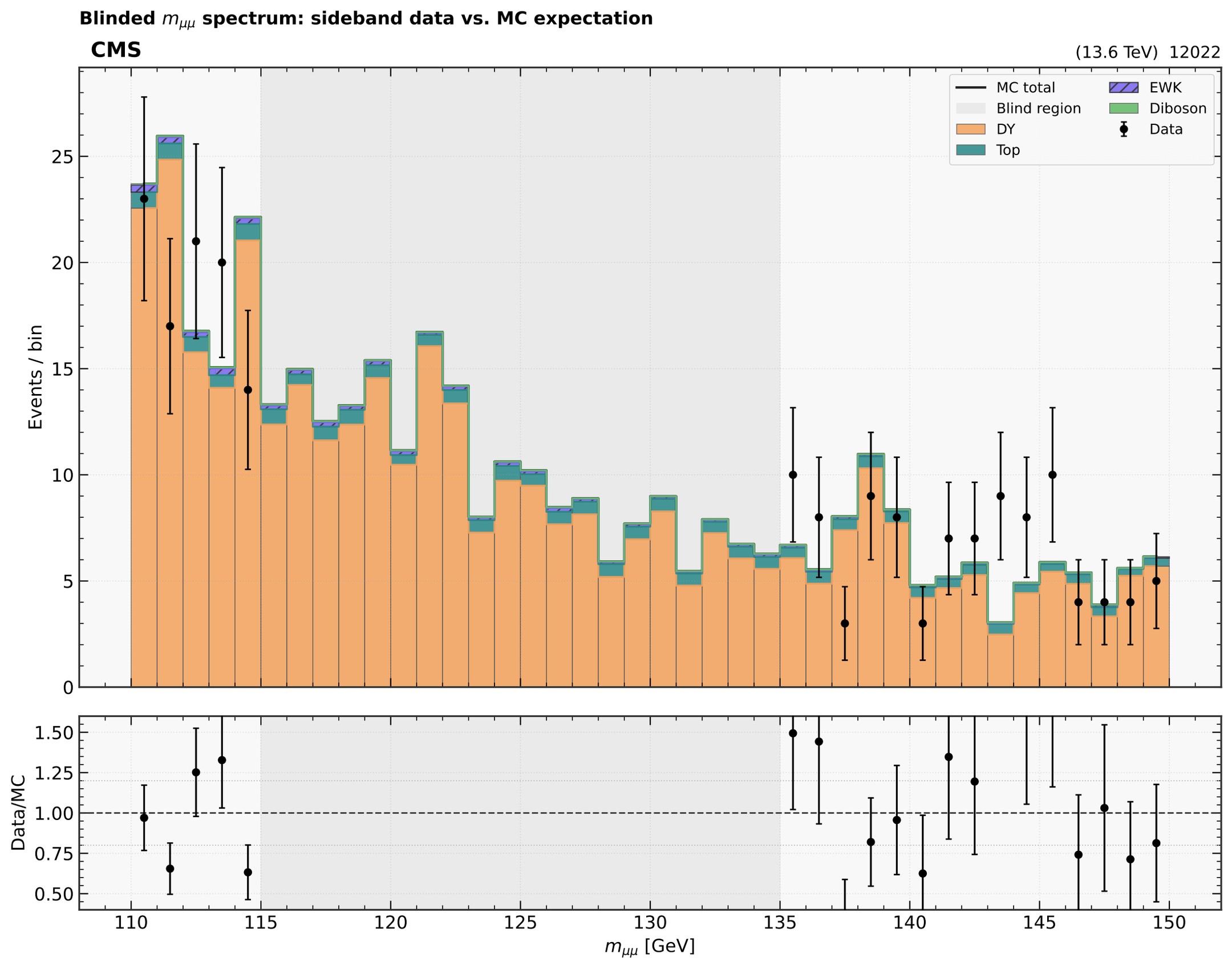



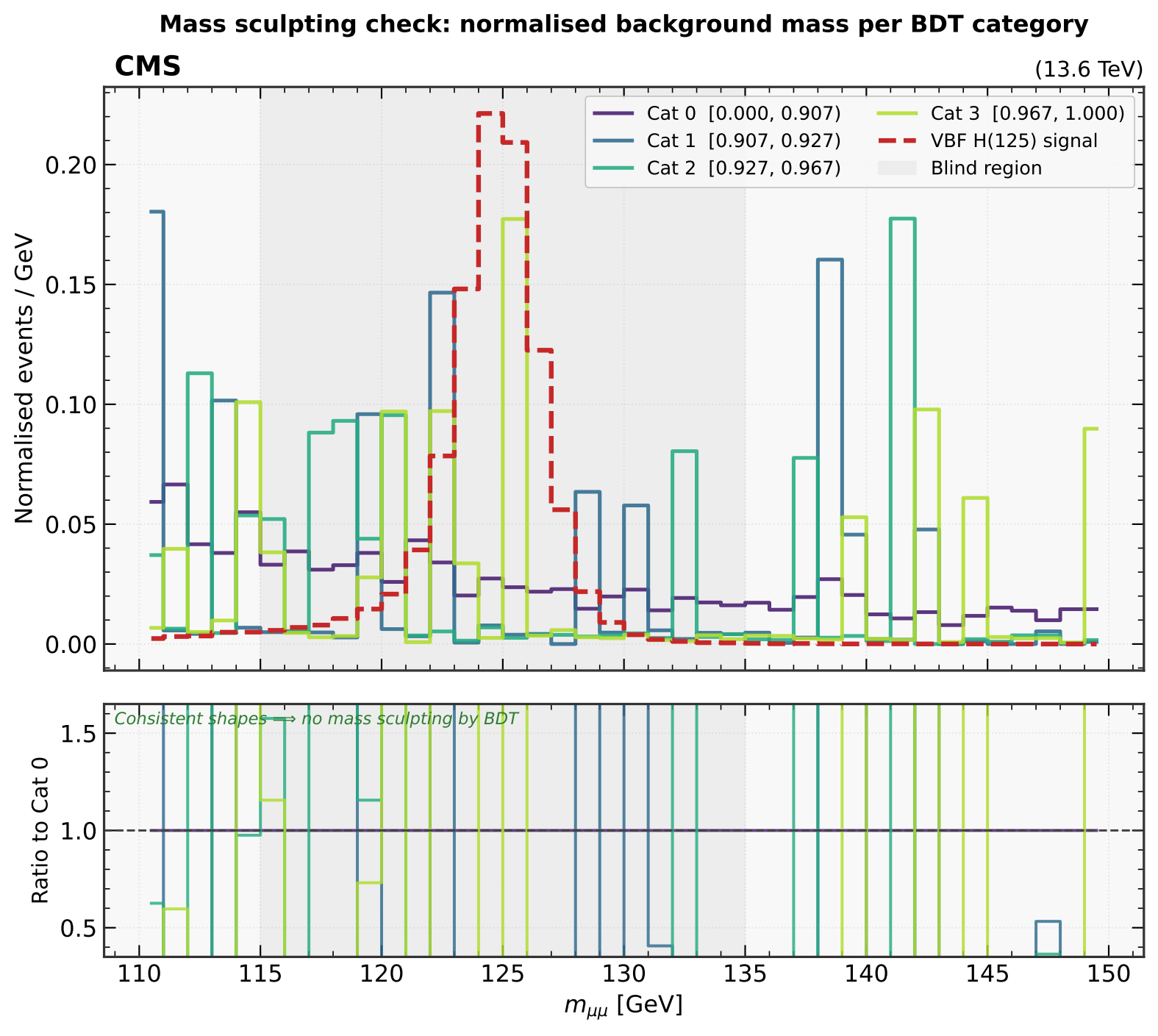
**VBF BDT score and per-bin $S/\sqrt{B}$ in signal HWHM window**



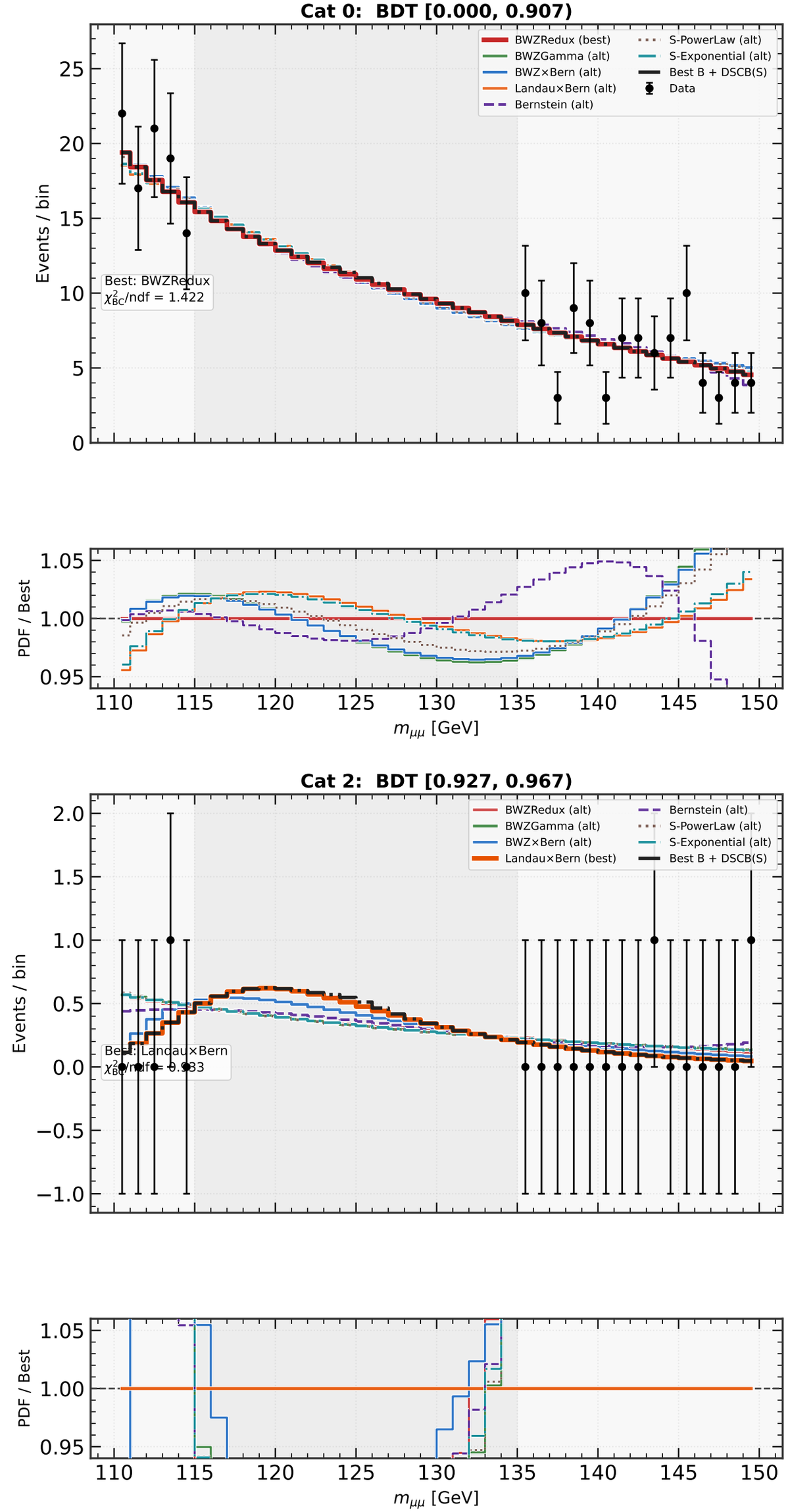

Dimuon mass resolution (FWHM) vs. VBF BDT score
CMS
(13.6 TeV)
Mean FWHM = 3.83 GeV
FWHM (2.355 × σ from DCB fit)
FWHM [GeV]
4.4
4.2
4.0
3.8
3.6
3.4
3.2
1.0
1.5
2.0
2.5
3.0
atanh((VBF BDT output + 1) / 2)

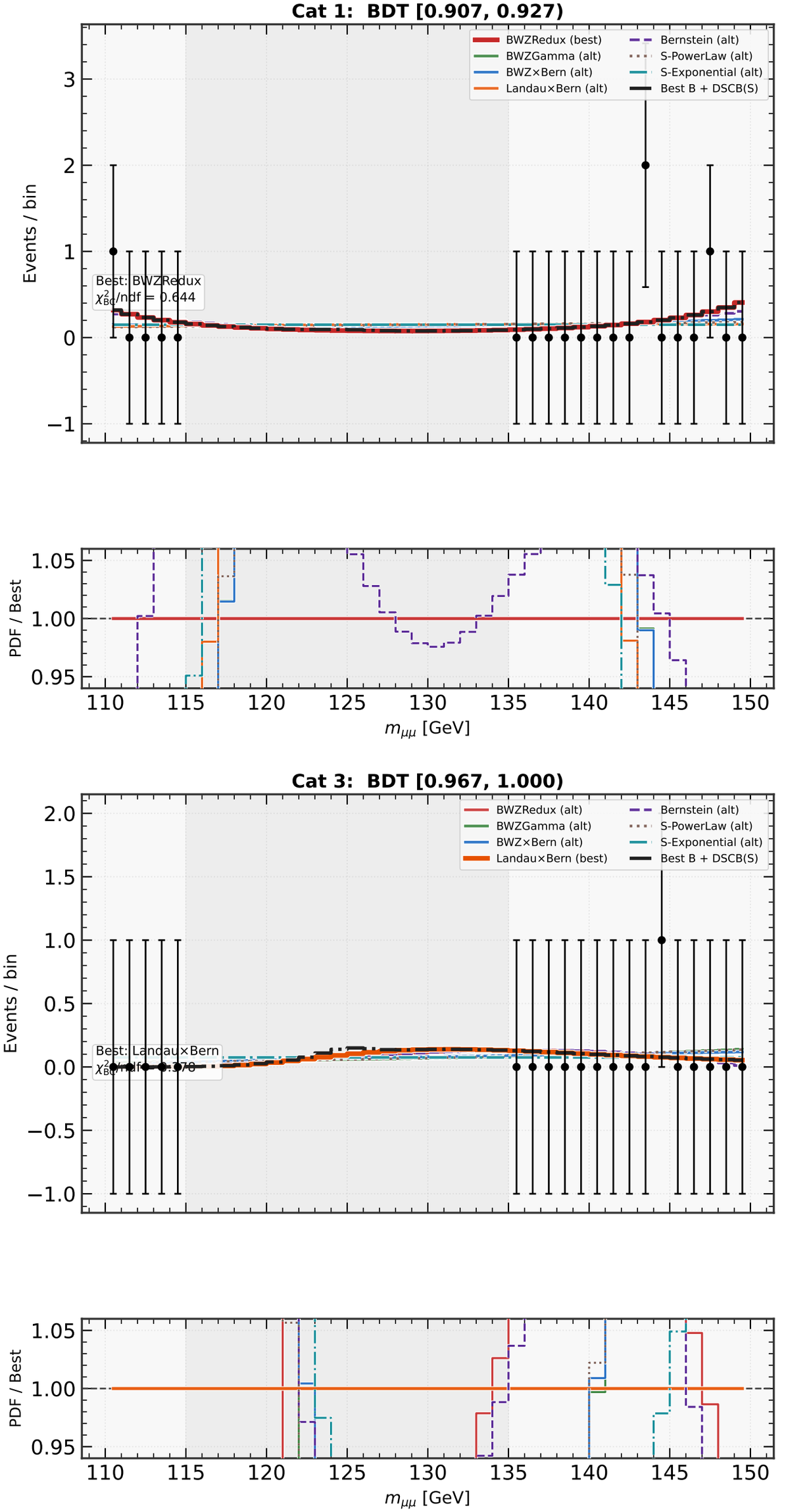

VBF BDT ROC curve
CMS
(13.6 TeV)
Signal efficiency
Background efficiency
0.0
0.2
0.4
0.6
0.8
1.0
Era 12022 (AUC = 0.882)


Blinded $m_{\mu\mu}$ spectrum: sideband data vs. MC expectation
CMS
(13.6 TeV) 12022
MC total
Blind region
DY
Top
EWK
Diboson
Data
Events / bin
Data/MC
$m_{\mu\mu}$ [GeV]


Mass sculpting check: normalised background mass per BDT category
CMS
(13.6 TeV)
Cat 0 [0.000, 0.907)
Cat 1 [0.907, 0.927)
Cat 2 [0.927, 0.967)
Cat 3 [0.967, 1.000)
VBF H(125) signal
Blind region
Normalised events / GeV
0.00
0.05
0.10
0.15
0.20
Consistent shapes ⟹ no mass sculpting by BDT
Ratio to Cat 0
0.5
1.0
1.5
110
115
120
125
130
135
140
145
150
$m_{\mu\mu}$ [GeV]

# Per-category dimuon mass: background model fits

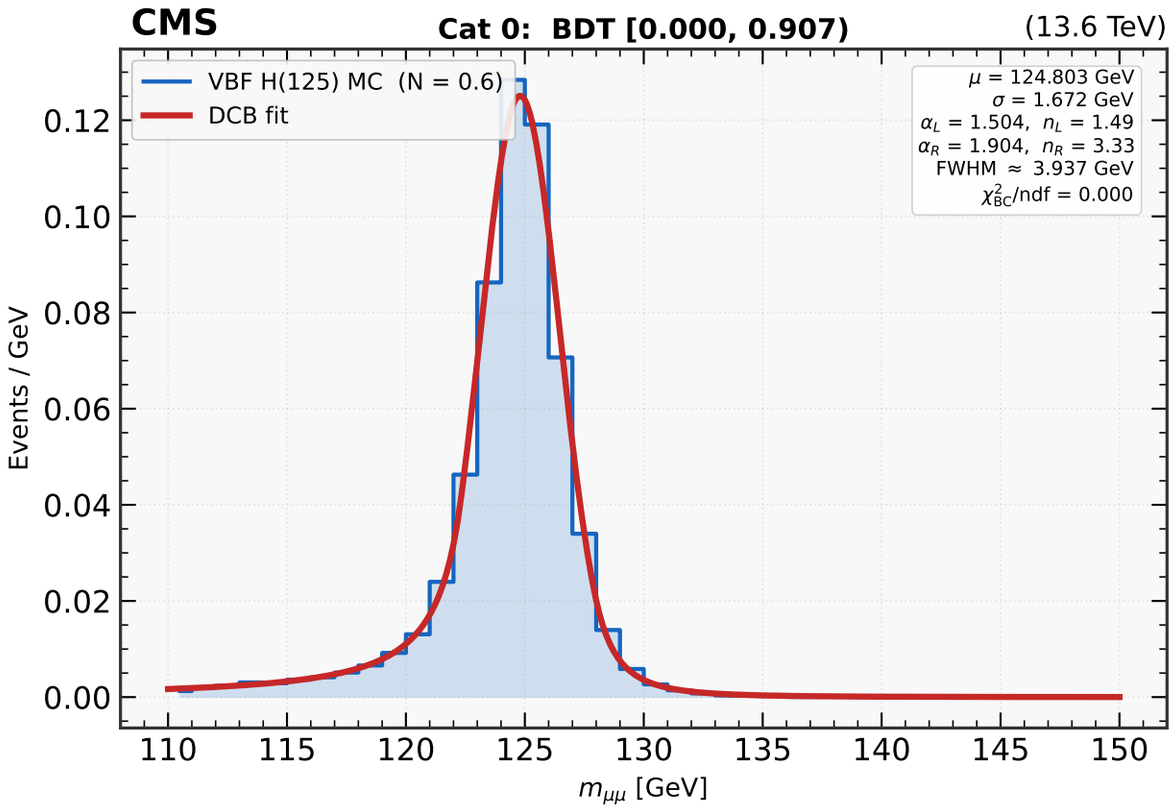


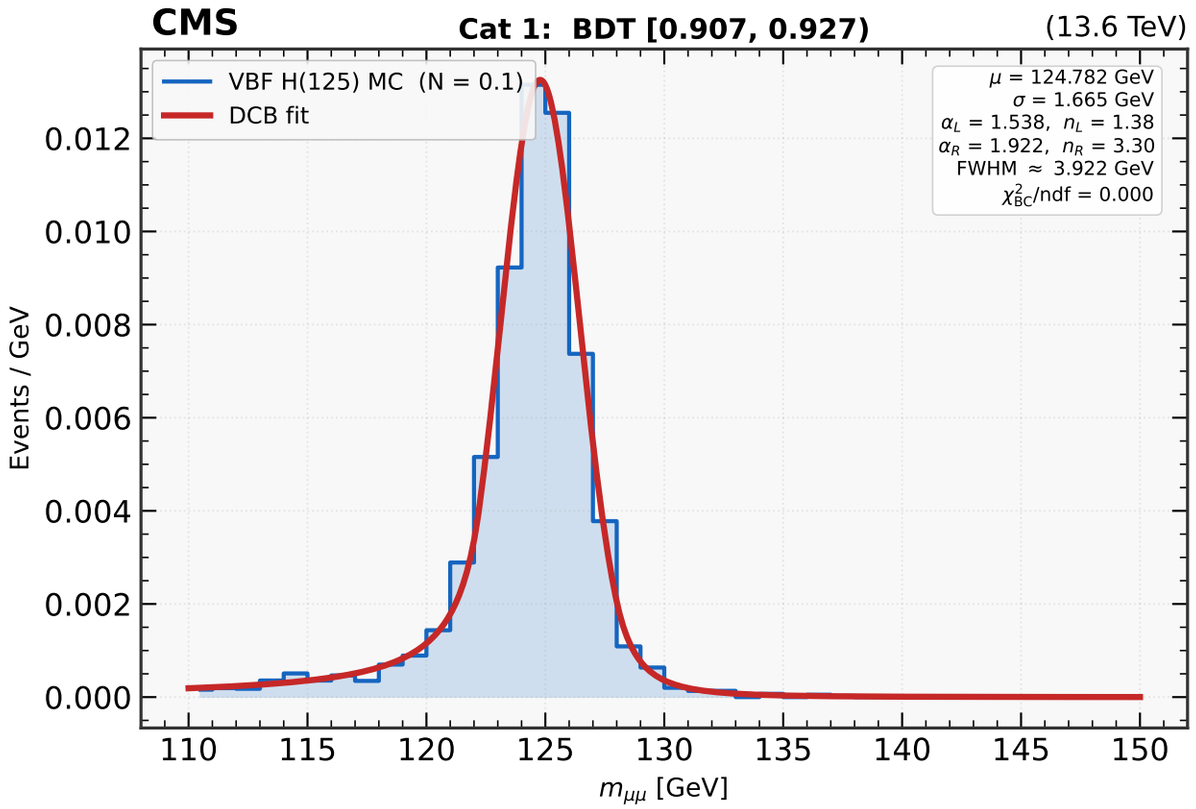



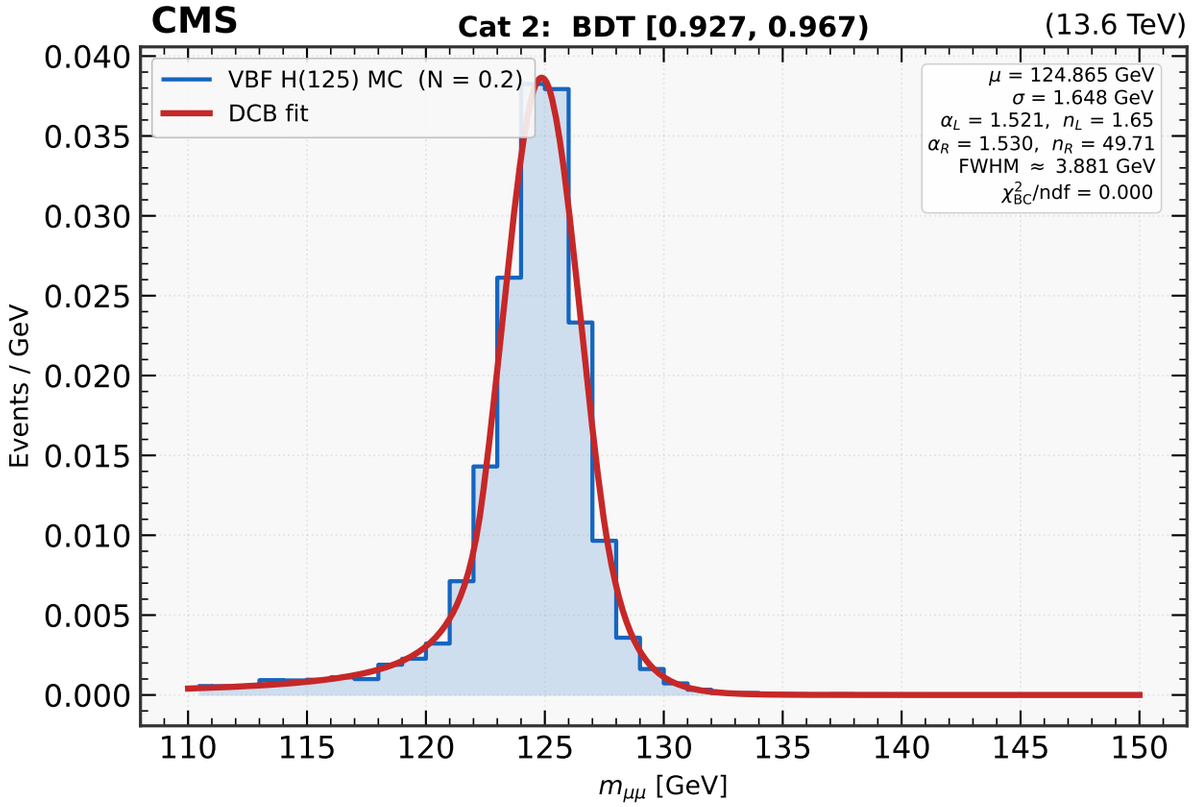
# Per-category background model fit comparison (χ²_BC/ndf, ndf)

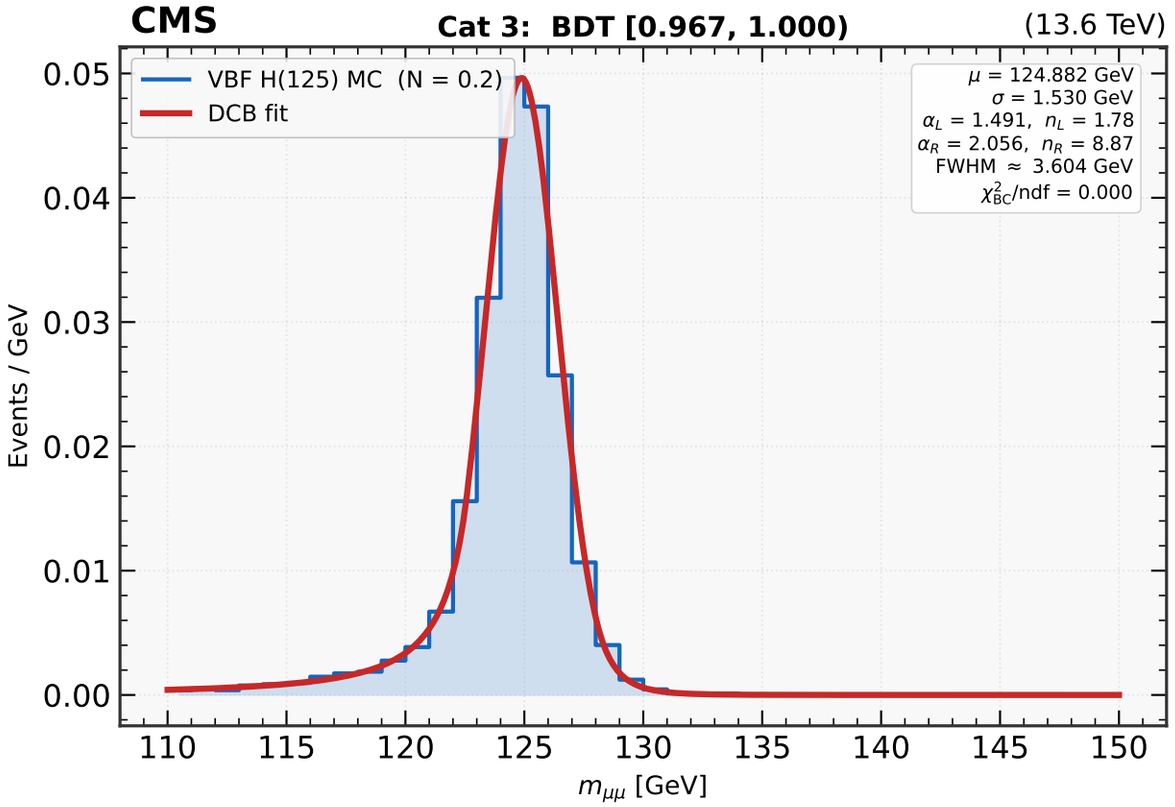
| Category | BDT range | Best model | Best χ²_BC/ndf | BWZRedux | BWZGamma | BWZ×Bern | Landau×Bern | Bernstein | S-PowerLaw | S-Exponential |
|---|---|---|---|---|---|---|---|---|---|---|
| 0 | [0.000, 0.907) | BWZRedux | 1.422 | 1.422 (37) | 1.446 (37) | 1.484 (36) | 1.441 (37) | 1.505 (35) | 1.441 (37) | 1.440 (37) |
| 1 | [0.907, 0.927) | BWZRedux | 0.644 | 0.644 (37) | 0.672 (37) | 0.692 (36) | 0.693 (37) | 0.678 (35) | 0.688 (37) | 0.690 (37) |
| 2 | [0.927, 0.967) | Landau×Bern | 0.933 | 0.984 (37) | 0.974 (37) | 0.946 (36) | 0.933 (37) | 0.995 (35) | 0.977 (37) | 0.976 (37) |
| 3 | [0.967, 1.000) | Landau×Bern | 0.378 | 0.386 (37) | 0.409 (37) | 0.408 (36) | 0.378 (37) | 0.400 (35) | 0.406 (37) | 0.420 (37) |

**VBF H(125) signal DCB extraction per BDT category**



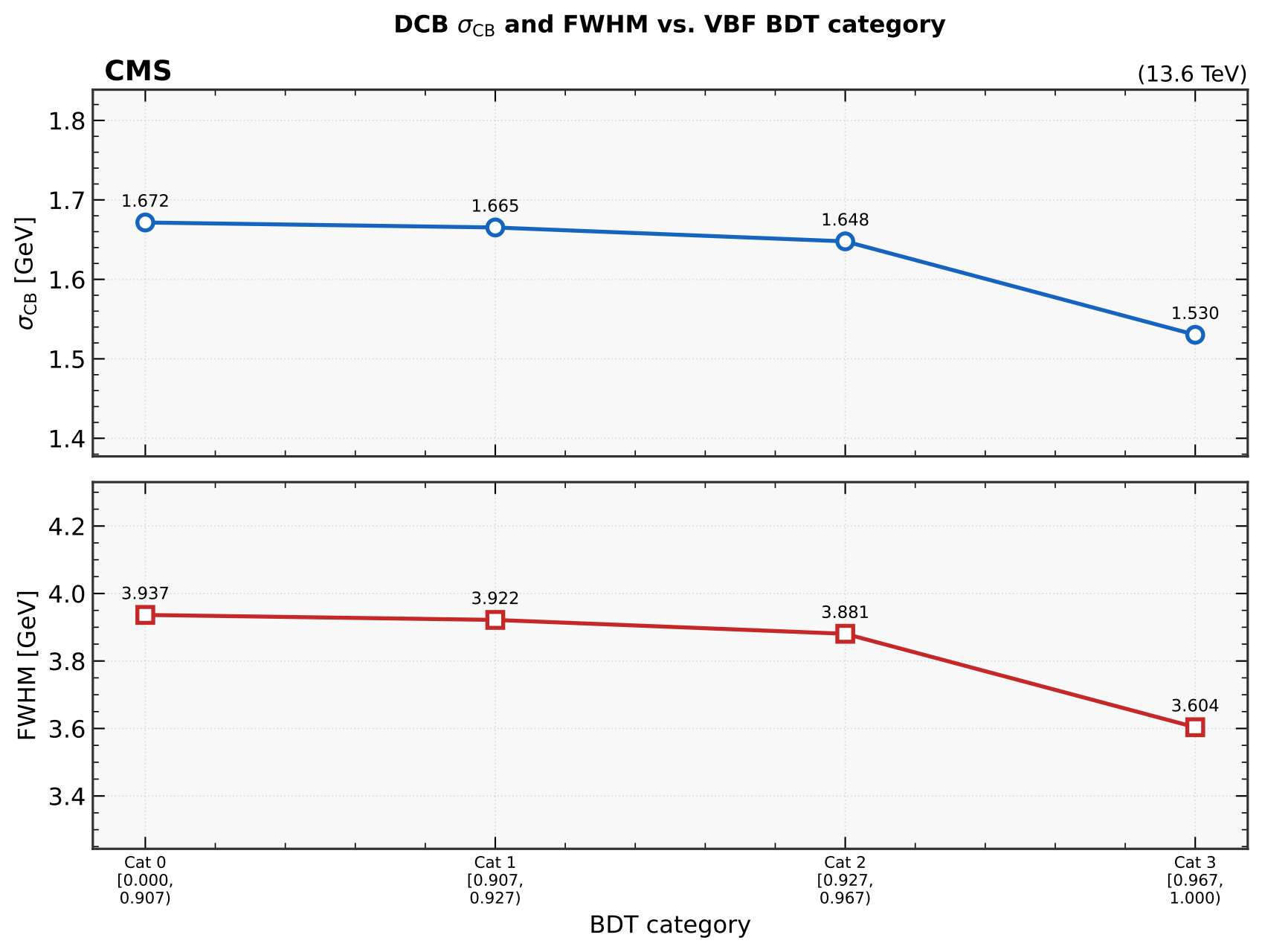



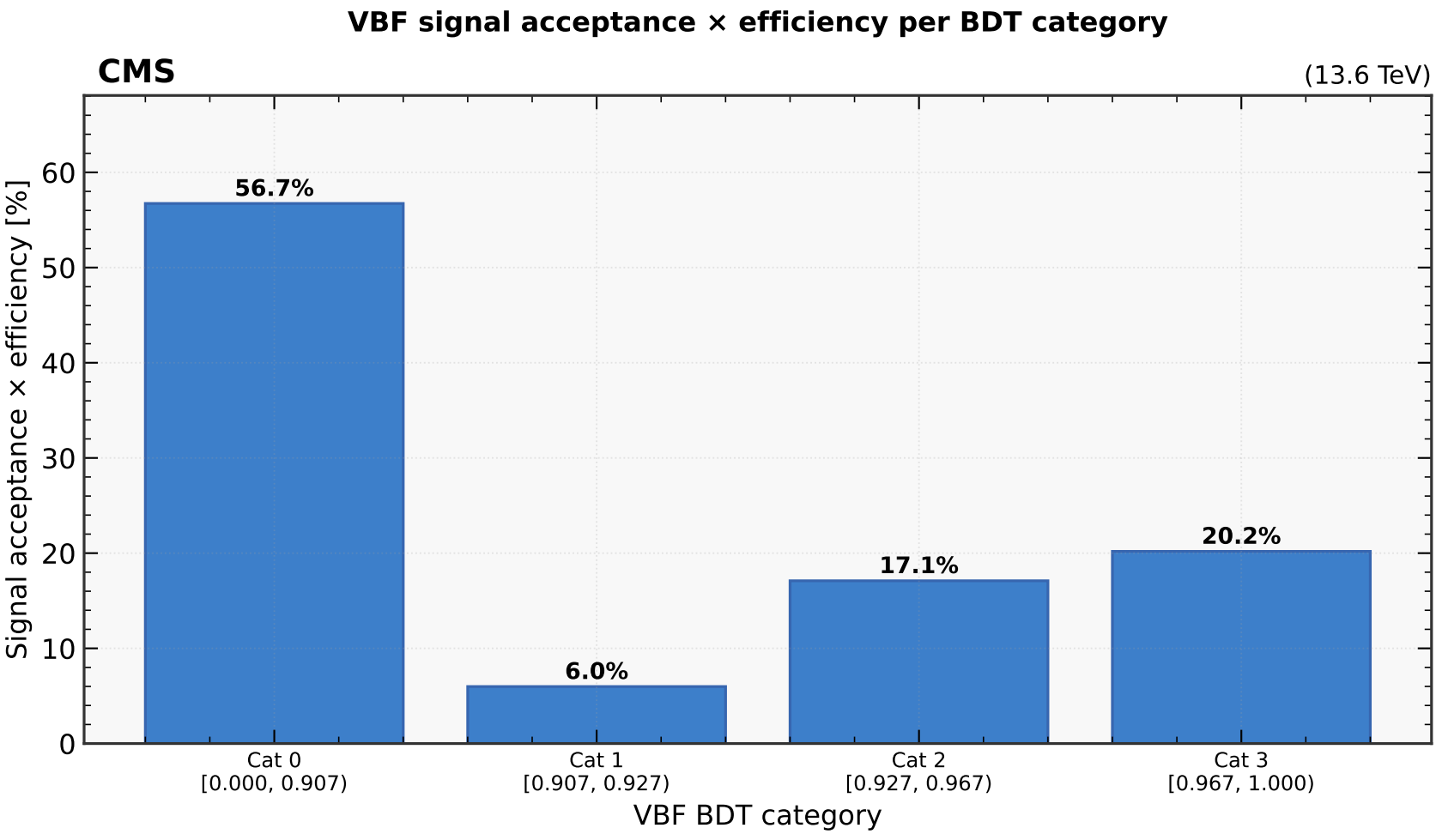



# VBF signal DCB fit parameters per category

| Category | BDT range | N_sig | Mean [GeV] | Sigma [GeV] | alpha_L | n_L | alpha_R | n_R | FWHM [GeV] | Asimov Z | χ²_BC/ndf |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 0 | [0.000, 0.907) | 0.59 | 124.8031 | 1.6716 | 1.5045 | 1.495 | 1.9036 | 3.325 | 3.937 | 0.038 | 0.000 |
| 1 | [0.907, 0.927) | 0.06 | 124.7819 | 1.6653 | 1.5379 | 1.384 | 1.9224 | 3.302 | 3.922 | 0.104 | 0.000 |
| 2 | [0.927, 0.967) | 0.18 | 124.8655 | 1.6478 | 1.5215 | 1.648 | 1.5301 | 49.712 | 3.881 | 0.224 | 0.000 |
| 3 | [0.967, 1.000) | 0.21 | 124.8825 | 1.5302 | 1.4907 | 1.784 | 2.0560 | 8.869 | 3.604 | 0.074 | 0.000 |


DCB $\sigma_{CB}$ and FWHM vs. VBF BDT category
CMS
(13.6 TeV)
$\sigma_{CB}$ [GeV]
1.8
1.7
1.6
1.5
1.4
1.672
1.665
1.648
1.530
FWHM [GeV]
4.2
4.0
3.8
3.6
3.4
3.937
3.922
3.881
3.604
Cat 0 [0.000, 0.907)
Cat 1 [0.907, 0.927)
Cat 2 [0.927, 0.967)
Cat 3 [0.967, 1.000)
BDT category


VBF signal acceptance × efficiency per BDT category
CMS
(13.6 TeV)
Signal acceptance × efficiency [%]
0
10
20
30
40
50
60
56.7%
6.0%
17.1%
20.2%
Cat 0
[0.000, 0.907)
Cat 1
[0.907, 0.927)
Cat 2
[0.927, 0.967)
Cat 3
[0.967, 1.000)
VBF BDT category

## Per-category yield and sensitivity summary

| Category | BDT range | S_full | B_full | S_HWHM | B_HWHM | HWHM window [GeV] | S/√B | Asimov Z |
|---|---|---|---|---|---|---|---|---|
| 0 | [0.000, 0.907) | 0.59 | 1208.62 | 0.402 | 110.545 | [122.8, 126.8] | 0.038 | 0.038 |
| 1 ★ | [0.907, 0.927) | 0.06 | 10.06 | 0.042 | 0.149 | [122.8, 126.7] | 0.109 | 0.104 |
| 2 ★ | [0.927, 0.967) | 0.18 | 18.69 | 0.123 | 0.266 | [122.9, 126.8] | 0.239 | 0.224 |
| 3 | [0.967, 1.000) | 0.21 | 18.40 | 0.147 | 3.964 | [123.1, 126.7] | 0.074 | 0.074 |

## Category summary: HWHM-windowed signal and background yields

| Category | BDT range | Data sidebands | MC sidebands ± stat | Data/MC (sidebands) | MC full (110-150) ± stat | S/√B [HWHM] | Asimov Z (cat) [HWHM] | B_HWHM |
|---|---|---|---|---|---|---|---|---|
| 0 | [0.000, 0.907) | 186 | 583.93 ± 26.39 | 0.319 | 1208.62 ± 37.93 | 0.0382 | 0.0382 | 110.545 |
| 1 ★ | [0.907, 0.927) | 4 | 5.75 ± 2.64 | 0.696 | 10.06 ± 3.26 | 0.1086 | 0.1041 | 0.149 |
| 2 ★ | [0.927, 0.967) | 3 | 9.21 ± 3.39 | 0.326 | 18.69 ± 4.84 | 0.2391 | 0.2235 | 0.266 |
| 3 | [0.967, 1.000) | 1 | 8.97 ± 3.40 | 0.111 | 18.40 ± 4.72 | 0.0740 | 0.0735 | 3.964 |
| Combined | all | | | | | | 0.2601 | |

## Category composition by process (full mass window)

| Category | BDT range | DY | EWK | Top | Diboson | total | total_err2 |
|---|---|---|---|---|---|---|---|
| 0 | [0.000, 0.907) | 1123.37 | 12.32 | 69.38 | 3.55 | 1208.62 | 1438.57 |
| 1 | [0.907, 0.927) | 8.78 | 0.96 | 0.28 | 0.04 | 10.06 | 10.60 |
| 2 | [0.927, 0.967) | 16.26 | 1.97 | 0.41 | 0.05 | 18.69 | 23.41 |
| 3 | [0.967, 1.000) | 16.14 | 2.03 | 0.22 | 0.01 | 18.40 | 22.28 |